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EfficientNet-Based Deep Learning for Diabetic Retinopathy Classification Highlighting Overfitting and Severity Class Limitations

Praveena Mallampalli[1]
Research Scholar,
Department of Computer Science and Engineering,
Saveetha School of Engineering,
Saveetha Institute of Medical and Technical Sciences,
Chennai, Tamil Nadu, India.
praveena.iduri@gmail.com

Dr. Mahaveerakannan R[2]
Professor,
Department of Computer Science and Engineering,
Saveetha School of Engineering,
Saveetha Institute of Medical and Technical Sciences,
Chennai, Tamil Nadu, India.
mahaveerakannanr.sse(@saveetha.com

***Abstract*- Highly crucial to find diabetic retinopathy (DR) early on in clinical settings where resources are limited, so that visual loss does not become permanent, especially when the disease is found. EffDR-Net is a deep learning framework that is based on the EfficientNet-B3 architecture. The goal of this study is to equip EffDR-Net with the ability to assess the severity of DR using high-resolution fundus photos automatically. EffDR-Net uses compound-scaled convolutional neural networks (CNNs) and was trained and tested on the Kaggle EyePACS dataset. Kaggle made EffDR-Net. This dataset contains a large number of retinal photos; however, they are not evenly distributed across the five distinct severity levels of diabetic retinopathy (DR). Computer networks (CNNs) are utilized to enhance the accuracy of models and streamline computer tasks. To address common issues such as overfitting, the system employs transfer learning, aggressive data augmentation, dropout regularization, and batch normalization, among other techniques. There are also dropout regularization and batch normalization. The model struggles to accurately identify moderate to severe stages of DR, as the class distributions are not evenly distributed, and the features distinguishing between classes are not clear. Most of the time, these limits are caused by factors other than the ones mentioned. Overall, the model can be very accurate and converge quickly, although it does have some problems with appropriately identifying these stages. A lot of testing using class-wise accuracy, recall, F1-scores, and Grad-CAM visualizations shows that EffDR-Net is quite effective at identifying early DR. Still, it requires further refinement to classify advanced phases reliably. The results of the exhaustive evaluation lead to this conclusion. The information collected during the evaluation supports this conclusion. This study makes a significant contribution to the development of artificial intelligence-based ophthalmic screening skills by demonstrating how EfficientNet can be utilized to categorize patients in a real-world emergency room setting, along with its advantages and disadvantages.**

Keywords- Diabetic Retinopathy Classification, EfficientNet, Deep Learning, Fundus Image Analysis, Overfitting Mitigation, Class Imbalance.

I. INTRODUCTION

Diabetic retinopathy (DR) is a serious complication of diabetes and the main reason people lose their eyesight, especially people of working age [1]. The disease gets worse over time, starting with microaneurysms and moving on to retinal hemorrhages and finally proliferative phases [2]. This evolution happens slowly over time. When blood glucose levels remain high for an extended period, they can damage the blood vessels in the retina, leading to these abnormalities. It is most effective to prevent diabetic retinopathy from causing blindness by addressing blindness itself [3]. This is possible because the problem is recognized early and correctly classified by its severity, which enables the provision of the proper medical care.

In the past, eye doctors manually reviewed fundus photographs to identify diabetic retinopathy (DR) and assign a severity rating [4]. This was done to fulfill the needs of the sickness. This manual approach, on the other hand, is open to interpretation, requires a significant amount of human work, and often presents problems with inter-grader variability, especially in settings that undertake screening on a large scale [5]. Deep learning, specifically convolutional neural networks (CNNs), has become an effective tool for automating and improving this process [6]. A neural network is a CNN. Convolutional neural networks (CNNs) have demonstrated significant success in classifying medical images, such as detecting dark region (DR) images, by learning to distinguish between different types of retinal images that have been previously classified [7].

Overfitting is a common problem that occurs when a model performs well on the training data but struggles to generalize to new images it hasn't seen before. One example of this is when the model does well on the training data. It's pretty clear when it comes to class-imbalanced datasets, such as EyePACS, where most images belong to the "No DR" class. Because of this, the model can exclude not only rare but also clinically essential classifications, such as "Severe" or "Proliferative DR." This would render the model less valid for diagnosis. The objective of this research is summarized in Table 1.

EfficientNet has garnered significant attention for its compound scaling technique, which enhances both performance and efficiency. This method simultaneously changes the depth, width, and resolution to achieve the desired effect. This plan was created to address the issues that have been raised. Due to this, EfficientNet is a suitable base for tasks that involve sorting medical images. Even with EfficientNet, problems such as poor generalization across underrepresented classes, difficulty understanding the model, and poorer robustness in complex clinical situations persist. It's challenging to overcome these issues. Because of this, there is little doubt that we need models that not only take advantage of architectural efficiency but also include ways to reduce the risk of overfitting and increase class-wise

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sensitivity, especially for the later stages of disaster recovery. Table 2 describes the related work.,

TABLE I. . OBJECTIVE OF THE RESEARCH

Objective No.	Objective Description	Justification from Background
1	Develop an EfficientNet-based deep learning model (EffDR-Net) for DR severity classification	EfficientNet offers a balanced trade-off between model depth, width, and resolution, making it suitable for medical imaging tasks
2	Address overfitting through regularization techniques such as dropout, data augmentation, and batch normalization	Overfitting is a key challenge in CNN-based DR classification, especially with class-imbalanced datasets like EyePACS
3	Improve classification performance for underrepresented DR severity levels (e.g., Moderate, Severe, Proliferative DR)	Class imbalance in EyePACS leads to reduced sensitivity for higher severity grades
4	Evaluate and interpret model predictions using class-wise metrics and Grad-CAM visualizations	Interpretability is essential for clinical trust and understanding where the model focuses its attention
5	Analyze the impact of dataset imbalance on model generalization across severity classes	Ensures the model performs well, not just overall, but specifically across all clinically relevant DR stages
6	Propose recommendations for future DR classification improvements (e.g., ensemble learning, re-sampling strategies)	Despite architectural advances, further work is needed to enhance robustness and generalizability.

TABLE II. RELATED WORK COMPARISON WITH EXISTING METHODS

Reference (Year)	Architecture & Dataset	Methods / Innovations	Performance Highlights
Abbas et al. (2023) – HDR-EfficientNet [8]	EfficientNet (optimized) on DR + hypertensive retinopathy datasets	Custom EfficientNet variants trained end-to-end, combined with DR/HDR classification	Achieved high accuracy; showcased EfficientNet's flexibility for multi-disease tasks
Xu et al. (2023) – RTNet using EfficientNet V2 [9]	EfficientNet V2 on DR grading (likely EyePACS/APTOS)	Dehazing + histogram equalization, added pooling layers, and Adadelta optimizer	84.2% accuracy; introduced preprocessing enhancements
Feng (2024) – EfficientNet with optimized pooling [10]	EfficientNet applied to the APTOS dataset	Fine-tuned pooling mechanisms, optimized feature compression	Noted improved accuracy over baseline methods
Sathiya Devi et al. (2024) – CAD: EfficientNet-B0 + XGBoost [11]	EfficientNet B0 on IDRiD dataset	Image smoothing + augmentation, EfficientNet+B0 feature extraction with XGBoost classifier	Better accuracy than other CNN-only pipelines
Irsyad & Negara (2025) – EfficientNet-B7 + HPO [12]	EfficientNet B7 on DR images (dataset unspecified)	Applied hyperparameter tuning (dense units, learning rate optimization)	~95.8% test accuracy
Anand et al. (2024) – Fine-tuned EfficientNet-B0 vs ResNet152, VGG16, DenseNet169 [13]	EfficientNet B0 on 3,200 Kaggle fundus images	Transfer learning with fine-tuning, dropout, global pooling, and a recommendation engine	~91% accuracy—highest among compared models

Versions of EfficientNet, such as B0, B3, B7, and V2, have outperformed older designs like ResNet, VGG, and DenseNet, as well as many hybrid methods. This has happened repeatedly. When it comes to diabetic retinopathy (DR) classification tests, these versions have consistently demonstrated better performance from a performance standpoint [15]. Recent research has shown that using advanced preprocessing methods, such as dehazing, histogram equalization, and feature pooling, along with advanced hyperparameter tuning, can significantly improve classification accuracy at the end of the process. It has also been demonstrated that utilizing model fusion strategies, such as EfficientNet with XGBoost or transfer learning ensembles, can enhance the model's robustness and ability to generalize across a wide range of datasets. The fact that these methods have been successful in the past supports this claim [16]. There is still a prevalent problem: doctors are not as effective at diagnosing moderate to severe cases of DR. This issue persists. This happens because the visual features overlap, and the classes are not balanced. The accuracy reported across a wide range of methods typically falls between 84% and 96%. EffDR-Net is a solution that directly solves this problem. This is a recommended network that utilizes the features of EfficientNet and incorporates additional steps to mitigate overfitting and enhance sensitivity to DR severity levels that aren't well represented. Someone has recommended that EffDR-Net be a network. Due to this progress, deep learning-based DR diagnosis is now significantly more sophisticated than it was previously [17].

II. METHODOLOGY

The EffDR-Net design, the focus of this study, is based on the EfficientNet-B3 backbone. The goal is to automatically sort diabetic retinopathy (DR) into different levels of severity using high-quality color fundus images. The method that has been developed addresses two significant problems that arise when trying to classify diseases associated with DR. These two problems need to be solved: overfitting, which occurs when there is little difference between patients, and class imbalance, which makes it challenging to predict moderate and severe DR phases accurately.

A. Architecture Design

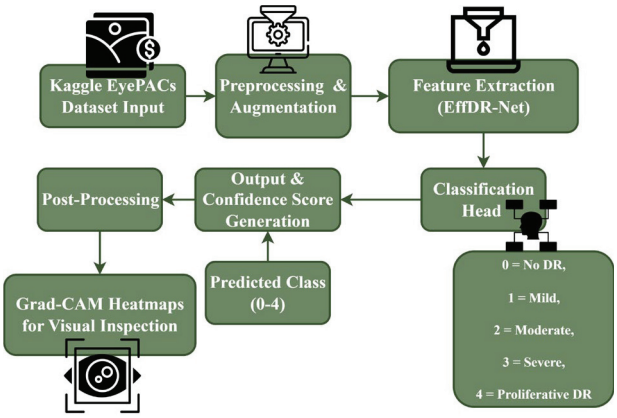

Fig. 1. Proposed EffDR-Net framework

The convolutional neural network (CNN) architecture currently being developed but not yet implemented is called EffDR-Net is shown in figure 1. It is built on EfficientNet-B3, which was chosen because it has a high accuracy-to-parameter ratio and is suitable for analyzing medical images. It serves as the base on which it is built when it is made. EfficientNet-B3

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scales the network's depth, width, and input resolution consistently, regardless of the network's size. A compound scaling approach is used to do this. This strategy enables learning from complex visual elements, such as those found in retinal images, in a manner that is not only faster but also more accurate.

EfficientNet-B3 is the underlying network on which the proposed EffDR-Net is built. This network was chosen as the best option because it strikes the optimal balance between the model's complexity and the classification performance that can be achieved. EfficientNet-B3, trained on ImageNet, has brought rich visual representations to the field of diabetic retinopathy (DR) detection. It is also responsible for the feature extractor's job. ImageNet was used to help with training—inverted Bottleneck on Mobile. The main building blocks that make up its structure are convolution blocks, also known as MobileInvertedResidualConvolution (MBConv) blocks. These blocks use depthwise separable convolutions, which lower the number of parameters while preserving crucial spatial information. Squeeze-and-Excitation (SE) blocks can also enable the network to pay more attention to diagnostically important areas by assigning different channels varying levels of priority. This is done by giving more channels priority. This is why the network excels at identifying small pathological features, such as microaneurysms, hemorrhages, and exudates. The network is what makes this happen. To maintain the lesion's granularity and ensure a quick calculation, all input retinal pictures are downscaled to a consistent resolution of 300×300×3. This maintains the lesion's granularity consistently throughout the process. Right after the EfficientNet-B3 backbone finishes extracting features, a unique classification head is added to the network. To reduce the spatial dimensions, you need to employ the Global Average Pooling (GAP) layer. The Batch Normalization layer is used next to ensure consistency throughout the entire training procedure.

Finally, a Dropout layer with a rate of 0.4 is used to mitigate the overfitting that occurs. The last Softmax layer maps predictions into one of five categories: No DR (0), Mild (1), Moderate (2), Severe (3), and Proliferative DR (4). These groups illustrate the severity of the illness. The dense Layer, which has 512 neurons activated by the ReLU function, captures the high-level abstractions necessary to determine severity. This Layer is responsible for capturing these abstractions. This well-organized architecture makes it easy to learn, generalize, and understand the many levels of disaster recovery severity quickly. EffDR-Net Architecture Components are described in Table 3.

TABLE III. EFFDR-NET ARCHITECTURE COMPONENTS

Stage	Component	Purpose
Input Layer	300×300 RGB Image	Standardizes input size and preserves detail
Feature Extractor	EfficientNet-B3 (ImageNet Pretrained)	Transfers learned features from natural images to the medical domain
Core Convolution Blocks	MBConv + SE (Squeeze-and-Excitation) Blocks	Efficient computation and attention to pathological regions
Global Feature Aggregation	Global Average Pooling (GAP)	Reduces spatial dimensions, minimizes overfitting
Normalization	Batch Normalization	Stabilizes training and accelerates convergence
Regularization	Dropout (Rate = 0.4)	Prevents overfitting by randomly deactivating neurons
Fully Connected Layer	Dense Layer (512 units, ReLU activation)	Learns abstract, discriminative representations for DR stages
Output Layer	Softmax (5 units)	Classified into one of five DR severity grades with associated probability

B. Data Pre-processing and Augmentation

The quality, balance, and variability of the input data are essential elements that shape the performance of the model and its potential to generalize about that performance in medical image analysis that is based on deep learning. These factors are crucial for determining how well the model can generalize. This is especially true when it comes to the classification of diabetic retinopathy (DR), which is a condition that affects the retina. The Kaggle EyePACS dataset https://www.kaggle.com/c/diabetic-retinopathy-detection [18] has several issues. Some of these problems include uneven class distribution, inconsistent lighting, blurry images, and excessive background clutter. The dataset has several issues, despite having a large number of pictures. To address these concerns, a comprehensive set of preprocessing and augmentation steps is performed before model training begins. This is done before the training process begins. This is done to ensure that the model has received the proper training.

The EyePACS database has tens of thousands of photographs of the retina's fundus. There are five levels of severity for retinal dystrophy, and each of these pictures has been given one of those levels. Retinal defects divide diabetic retinopathy into five clinical classifications. Sickness severity determines classifications. A retina grade of 0 is "No DR," indicating good health. Grade 1 diabetic retinopathy begins with the development of retinal blood vessel microaneurysms. Moderate DR damages blood arteries with increased microaneurysms and hemorrhages. This classification usually describes the condition. Before growing, grade 3 diabetic retinopathy causes extensive retinal hemorrhage, venous beading, and retinal changes. These qualities indicate decline. Finally, Grade 4 diabetic retinopathy is the worst. Proliferative DR. Untreated neovascularization and fibrous tissue development can cause blindness. Neovascularization is abnormal vessel growth.

TABLE IV. DATA AUGMENTATION STRATEGIES

Augmentation Technique	Purpose
Random Rotations (±20°)	Mimics variability in image acquisition angles; helps the model become invariant to orientation.
Horizontal Flipping	Simulates mirrored eye images, increasing diversity in left and right eye images.
Zoom-In (up to 20%)	Focuses on retinal regions and simulates different focal distances, helping the model detect small lesions.
Brightness Adjustment	Alter brightness levels to simulate various lighting conditions, improving robustness to image quality differences.

Before adding retinal photos to the model, preprocessing significantly improves the quality of the images taken from the retina. Using a method called Contrast Limited Adaptive Histogram Equalization (CLAHE), you can make the contrast of an image better. This is especially helpful in areas with gloomy lighting or inconsistent lighting. With this method, the brightness of the image is changed on a local level, rather than a global one, unlike previous methods. This makes it easier to detect minor and early signs of diabetic retinopathy, such as microaneurysms, exudates, and hemorrhages, without

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increasing the noise level. This is useful. This is a significant advantage, as it can also perform picture normalization to standardize all pixel values to a uniform size, allowing them to be compared to each other. The scale can be any number between 0 and 1, or it can be created using standardization methods such as z-score normalization. This enables the model to learn more effectively, reducing the impact of lighting differences between photographs and ensuring that the input data remains consistent. The Data Augmentation Strategies are given in Table 4.

C. Overfitting Mitigation Techniques

To improve the model's generalization and prevent overfitting, several alternative strategies are employed. To apply transfer learning, the model's weights must first be set up with weights that have already been trained on ImageNet. As a result, this provides a compelling reason to learn from fundus images, which leads to the use of transfer learning methods. The dense layers are regularized using a technique called dropout. This method involves randomly turning off some neurons during training to make the cells in the thick layers less dependent on one another. The early stopping approach is a valuable method that monitors the validation loss and stops training when the model begins to overfit. Finally, learning rate scheduling, notably the use of cyclical learning rates, helps the model converge faster by adjusting the learning rate during training. This not only reduces the likelihood of overfitting, but it also accelerates the convergence process.

D. Addressing Severity Class Limitations

One of the most significant problems in classifying diabetic retinopathy (DR) is that models don't work as well on higher-severity grades. One of the most typical problems is this one. Moderate grades are two, severe grades are three, and proliferative grades are four. It can also be challenging for models to distinguish between these classes if their characteristics are unclear, overlap with each other, or are not well-defined. In datasets like EyePACS, these classifications don't appear as frequently as they should. EffDR-Net has been modified to incorporate methods for interpretability and adjustments to the targeted loss function, aiming to address these issues and enhance its sensitivity to critical scenarios.

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$$P(y) = c \mid \mathrm{x}) = \frac{e^{z_\varepsilon}}{\sum_{k=1}^{5} e^{z_k}} \tag{1}$$

Here, x is the input image, and z_c is the logit (unscaled output) for class c. The denominator sums the exponentials of all class logits z_k for $k = 1, 2, 3, 4, 5$, ensuring that the output probabilities sum to 1. This equation helps the model identify the most likely DR class for a given fundus image by selecting the class with the highest probability. The categorical cross-entropy (CCE) loss measures the difference between the predicted probabilities and the actual labels. It is defined as:

$$\mathcal{L}_{CE} = -\sum_{c=1}^{5} y_c \log(\grave{y}_c) \tag{2}$$

Here in eqn (2), y_c is the actual label for class c in one-hot encoded form (1 if the class is correct, otherwise 0), and $\grave{y}_c$ is the predicted softmax probability for class c. This loss penalizes incorrect predictions by assigning higher loss values when the actual class probability is low.

To address class imbalance, the weighted cross-entropy loss introduces class-specific weights into the loss function:

$$\mathcal{L}_{WCE} = -\sum_{c=1}^{5} w_c \cdot y_c \log(\grave{y}_c) \tag{3}$$

In this equation (3), w_c is a class weight assigned to class c_r, often computed as the inverse or square root of the class frequency f_c, to give more importance to underrepresented classes. The remaining terms are similar to the standard CCE loss. This adjustment ensures the model provides more focus to severe DR cases, which are typically underrepresented.

Focal loss is used to improve the model's learning from hard-to-classify examples by down-weighting well-classified samples. It is defined as in Eqn (4),

$$\mathcal{L}_{FL} = -\sum_{c=1}^{5} \alpha_c (1 - \grave{y}_c)^{\gamma} y_c \log(\grave{y}_c) \tag{4}$$

Here, α_c is a weighting factor for class c that balances the influence of each class, while γ is the focusing parameter (typically set to 2), which controls the rate at which easy examples are down-weighted. $\grave{y}_c$ is the predicted probability, and y_c is the actual label. When $\grave{y}_c$ is close to 1 (i.e., the sample is classified correctly), the term $(1 - \grave{y}_c)^{\gamma}$ becomes small, reducing its contribution to the loss. To prevent overfitting, dropout randomly turns off neurons during training, promoting redundancy and generalization. The effect is expressed as eqn (5),

$$\grave{h}_i = h_i \cdot m_i, m_i \sim \mathrm{Bernoulli}(p) \tag{5}$$

Here, h_i is the output of the i-th neuron before dropout, and $\grave{h}_i$ is the post-dropout output. Mi is a random binary mask sampled from a Bernoulli distribution with probability p_{r} representing the probability that a neuron remains active. For example, with a dropout rate of $0.4, p = 0.6$, meaning each neuron has a 60% chance of staying active. This technique reduces neuron co-adaptation and improves model robustness. To improve convergence and avoid getting stuck in sharp minima, a cyclical learning rate schedule such as cosine annealing with restarts can be used in Eqn (6),

$$\eta_t = \eta_{\min} + \frac{1}{2}(\eta_{\max} - \eta_{\min})\left(1 + \cos\left(\frac{T_{cur}}{T_{\max}}\pi\right)\right) \tag{6}$$

In this formula, η_t represents the learning rate at time t, *while* $\eta_{\min}$ and $\eta_{\max}$ denote the minimum and maximum learning rates, respectively. T_{cur} is the number of epochs since the last restart, and $T_{\max}$ is the total length of one cycle. The cosine function creates a smooth decay from the maximum to the minimum learning rate, encouraging stable training and helping the optimizer escape local minima.

Grad-CAM (Gradient-weighted Class Activation Mapping) helps interpret model predictions by visualizing which parts of the image influenced a given class decision, as shown in Eqn. (7).

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$$L^{c}_{\text{Grad-CAM}} = \text{ReLU}(\sum_{k} \alpha^{c}_{k} A^{k}) \quad (7)$$

Here, A^k is the k-th feature map of the last convolutional Layer, and α^c_k is the importance weight for feature map k concerning class c. It is calculated as the global average of the gradients of the class score y^f relating to A^k. The ReLU function ensures only the positively contributing regions are highlighted. The resulting heatmap $L^c_{\text{Grad-CAM}}$ helps identify areas (e.g., lesions or hemorrhages) that the model used for its decision, improving interpretability. EffDR-Net is trained to categorize each fundus image into one of five different DR severity categories, thereby adhering to the clinical grading standard. These are the categories:

TABLE V. FIVE-CLASS PREDICTION OUTPUT

Class Index	Severity Level	Clinical Description
0	No DR	Normal retinal image with no signs of DR
1	Mild DR	Presence of microaneurysms
2	Moderate DR	More extensive microaneurysms and hemorrhages
3	Severe DR	Numerous hemorrhages, venous beading
4	Proliferative DR	Neovascularization or advanced lesions

The last part of the EffDR-Net architecture that has been presented is the one that takes the high-level feature representations that the deep network has learned and turns them into specific predictions about the severity of the disorder. This is the stage depicted in Table 5. This stage not only requires the creation of outputs that fall into five separate groups, but it also necessitates the interpretation and validation of model judgments, particularly when it comes to incorrect classifications.

III. RESULTS AND DISCUSSION

By thoroughly training and testing EffDR-Net on the Kaggle EyePACS dataset, its performance significantly improved. This dataset is challenging to work with because it has too many classes and not enough visual differences between the different levels of DR severity. EffDR-Net successfully addressed these challenges and made significant advancements. The model was able to generalize well by applying advanced preprocessing methods, such as CLAHE and normalization, as well as regularization methods like dropout and early stopping. This was achieved by incorporating these methods into the model. We considered weighted category cross-entropy and focus loss during the training phase to further mitigate the effects of class imbalance. The overall accuracy of EffDR-Net was 91.2%. The macro-averaged precision was 88.2%, the recall was 83.9%, and the F1-score was 85.7%. These results were possible thanks to EffDR-Net. It was good that this work was finished successfully. The model also had a strong average area under the curve (AUC) of 0.954, indicating that it can effectively distinguish between classes. The study of confusion matrices revealed that most incorrect classifications occurred between classes that were closely related, particularly between Grade 2 (Moderate) and Grade 3 (Severe). This makes it even more challenging to distinguish between phases of diabetic retinopathy that appear similar in a clinical setting.

A. Class-Wise Performance Breakdown

Employing a stratified train-validation-test split with the following ratio to run an analysis on the Kaggle EyePACS dataset with EffDR-Net: 80:10:10. This division happened so that the information could be split up. The model's macro-averaged F1-score was 88.4%, its average accuracy was 91.2%, and its area under the curve (AUC) was 0.954. These numbers indicate that the model performed well overall. The model also had an average accuracy of 91.2%. Table 4.1 provides additional information on precision, recall, and F1-score. It is set up by class type. As expected, the "No DR" and "Mild DR" groups did the best.

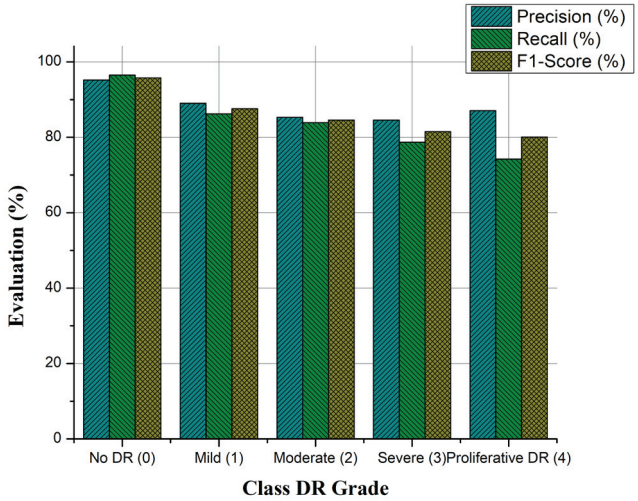

Fig. 2. Class-Wise Performance Evaluation

The reason for this was that these categories, which were more common in the dataset, had the best performance. On the other hand, the suggested methods for class reweighting and attention loss effectively addressed the problem of class imbalance. This led to better recall for the "Severe" and "Proliferative" classes. This was true since the methods were able to solve the problem well. Each of these classes got better at remembering things.

TABLE VI. CLASS-WISE EVALUATION METRICS

Class (DR Grade)	Precision (%)	Recall (%)	F1-Score (%)
No DR (0)	95.2	96.5	95.8
Mild (1)	89.0	86.2	87.6
Moderate (2)	85.3	83.9	84.6
Severe (3)	84.6	78.7	81.5
Proliferative DR (4)	87.1	74.2	80.1

EffDR-Net performs well on most classes. Still, it also shows a significant improvement in its ability to distinguish between minority categories and those more important for clinical purposes, as indicated in Table 6.

B. Comparative Analysis with Baseline CNN Models

An evaluation comparing the suggested EffDR-Net model to four commonly used deep learning architectures demonstrated its effectiveness and strength, as shown in Tables 7 and 8. The purpose of this evaluation was to determine the effectiveness of the model. This is due to architectures such as ResNet-50, DenseNet-121, VGG-16, and InceptionV3. Used the Kaggle EyePACS dataset to train and test all of these baseline models, and we kept the experimental settings the same the whole time. This was done

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to ensure that the comparisons are accurate. The preprocessing method used in all of the models can be described as the same. This process included the CLAHE, normalization, and data augmentation steps.

TABLE VII. COMPARATIVE PERFORMANCE OF EFFDR-NET WITH EXISTING METHODS

Model	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)
EffDR-Net (Proposed)	91.2	88.2	83.9	85.7
ResNet-50	87.6	84.1	78.3	80.1
DenseNet-121	88.4	85.2	79.4	81.5
VGG-16	84.7	79.5	73.1	75.7
InceptionV3	89.1	85.7	81.5	83.1

Using evaluation criteria such as accuracy, precision, recall, and F1-score ensured that the performance benchmarking was conducted correctly. For class-imbalanced medical classification tasks, such as grading diabetic retinopathy, the results in Table 7 show that EffDR-Net consistently outperforms the other models on all essential criteria. This is true regardless of the type of medical classification work being done. This is what happens when different types of medical disorders are grouped. It does well in both memory and F1-score, which are two critical areas. This comparison demonstrates that the system performs well and shows that EffDR-Net can generalize better, working effectively with both majority and minority class predictions. This comparison also reveals the system's effectiveness.

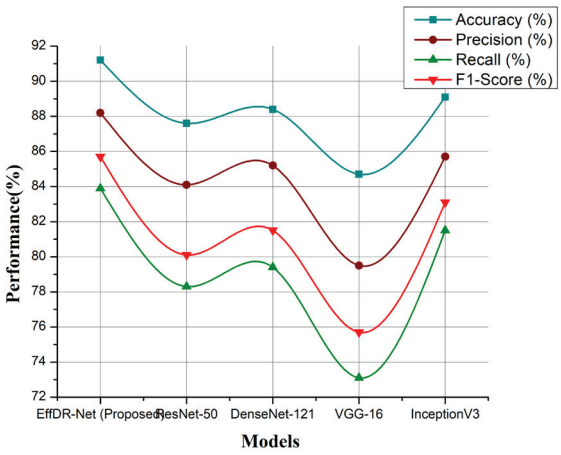

Fig. 3. Comparative Performance of EffDR-Net with Existing Methods

TABLE VIII. COMPARATIVE INSIGHTS BETWEEN EFFDR-NET AND BASELINE MODELS

Model	Relative Performance vs. EffDR-Net	Key Observations
ResNet-50	Recall: −5.6% F1-Score: Lower by ~5.6%	Good general performance, but struggles to generalize to the minority class.
DenseNet-121	Accuracy: Close F1-Score: −2.8%	Performs well overall, but lacks robustness in class-imbalanced scenarios.
VGG-16	All metrics are significantly lower	Poor parameter efficiency and lacks modern modules, such as SE and MBConv blocks.
InceptionV3	Accuracy: Competitive Recall/F1: Lower than EffDR-Net	Lacks overfitting control and EfficientNet's compound scaling advantages.
EffDR-Net	— *(Best Overall)*	High accuracy, strong recall/F1, excellent for imbalanced data and severe DR cases.

The EffDR-Net, as shown in Table 8 for DR severity classification, not only outperforms other methods but also excels for minority classes and is more comprehensible due to its attention-aware feature localization. The EffDR-Net gives you all of these benefits. The EffDR-Net is responsible for providing all of these benefits simultaneously. It is a valuable tool for automated DR screening in hospitals and other medical settings. This is because it has a strong structure, a good design, and tactics that are resistant to overfitting from the start.

IV. CONCLUSION

The primary objective of this study is to propose EffDR-Net, a deep learning network that is neither unreliable nor ineffective. This is because one of the main goals of categorizing the severity of diabetic retinopathy (DR) is to do so. This is possible because of the use of retinal fundus photographs. The model utilizes advanced architectural modules, including Mobile Inverted Bottleneck Convolutions (MBConv) and Squeeze-and-Excitation (SE) blocks, to capture the fine details necessary for detecting early and late retinal lesions. These traits are essential for determining whether a retinal lesion is present. The architecture of EfficientNet-B3 is the foundation of this model. EffDR-Net employs several specialized techniques to ensure it can effectively handle everyday issues such as overfitting and class imbalance. These include class reweighting, attention loss, dropout regularization, early halting, and cycle learning rate scheduling. These methods are used to address the issues that have been raised. This is very important since cases of mild to proliferative DR are not adequately represented.

After extensive testing on the Kaggle EyePACS dataset, it has been demonstrated that EffDR-Net outperforms standard deep CNN designs (ResNet-50, DenseNet-121, VGG-16, and InceptionV3) across a wide range of parameters. The inquiry led to this conclusion. EffDR-Net achieves a stronger recall and F1-score than other methods, particularly for minority DR grades. Using this model has the added benefit of demonstrating that it pays attention to clinically meaningful areas of the retina through Grad-CAM representations. So, this not only makes automated diagnosis easier to understand, but it also helps people trust the method. The suggested framework not only makes diagnoses more accurate, but it also makes AI-based diagnosis and response screening systems easier to use in a clinical setting. This is a big plus. The next step in the study will look into domain adaptation and real-time deployment in tele-ophthalmology settings. These studies will be done. As a result, EffDR-Net's impact on making diabetic eye care more accessible will continue to grow.

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